Rolling-membrane actuator

Spider-inspired pneumatic joint for legged robots, designed, prototyped and tested at KU Leuven

KU Leuven and Flanders Make. Main researcher, manuscript in preparation for Advanced Science

Mirco Paternuosto

Junior Mechanical engineer, MSc KU Leuven

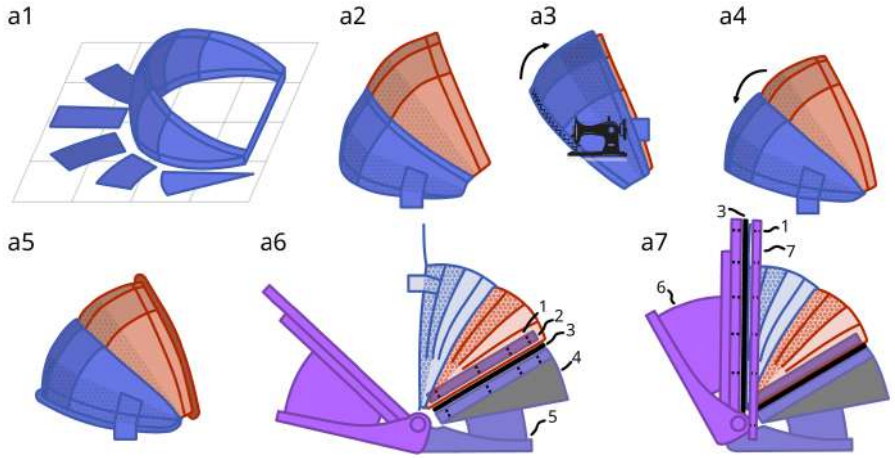
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Prototype fabrication

Patterned fabric, silicone bonding, stitching; modular 3D-printed clamps swap membranes fast.



Clamp, panels a6 and a7: **1** Bolts **2** Inner clamp **3** Neoprene gasket **4** Protrusion **5** Femur structure **6** Patella structure **7** Outer clamp

Result, prototype P8

Peak torque	**22.7 Nm** at 100 kPa
Torque factor	**0.35 Nm/kPa** average
Membrane	**33 µm** silicone-coated ripstop nylon
Durability	**1000 cycles** at 110°
Prototypes	**8** P1 to P8 compared

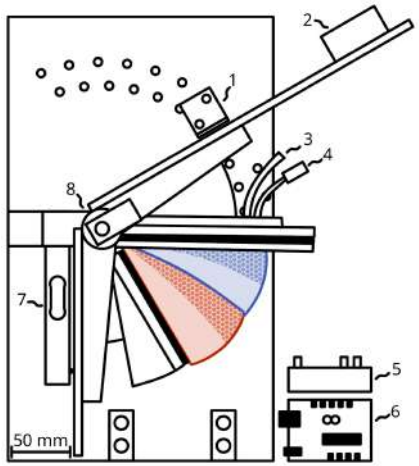
My part: custom test rig, control PCB and Arduino electronics, prototype iterations, Python and OpenCV analysis, failure root causes.

Workshop: Additive and subtractive fabrication, CNC machining, laser cutting, workshop tools

Workshop

Custom test rig

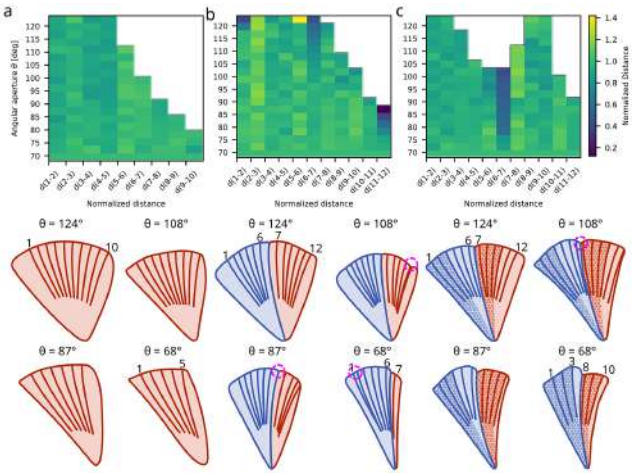
Control PCB and Arduino log torque, angle and pressure at 50 Hz.



1. Adjustable stopper
2. Loading lever
3. Air inlet
4. Pressure sensor
5. 3/2-way valve
6. Arduino control unit
7. Load cell
8. Potentiometer

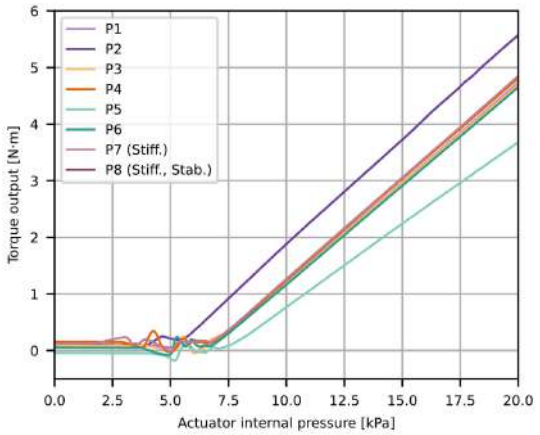
Failure mode analysis

Marker tracking separates stable rolling from collapse.



Torque against pressure

Eight prototypes, P1 to P8.



Composite layup optimisation

Surrogate-based optimisation of a stiffened fuselage panel, with open-source finite-element tools

Internship at Cenaero, Belgium. Aeronautical research center - March 2025 to July 2025

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Junior Mechanical engineer, MSc KU Leuven

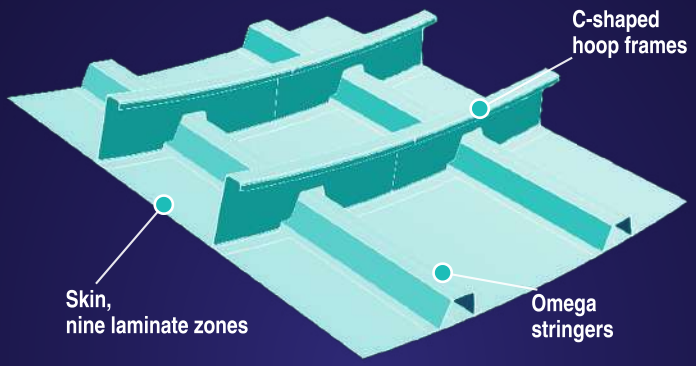
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Benchmark structure

Generic fuselage section, 1 m long: nine skin zones, two omega stringers and two C-shaped hoop frames.



Optimum layup

Plies per skin zone after 100 iterations on the enriched model.

z7 12	z8 12	z9 12
z4 14	z5 14	z6 14
z1 12	z2 12	z3 12

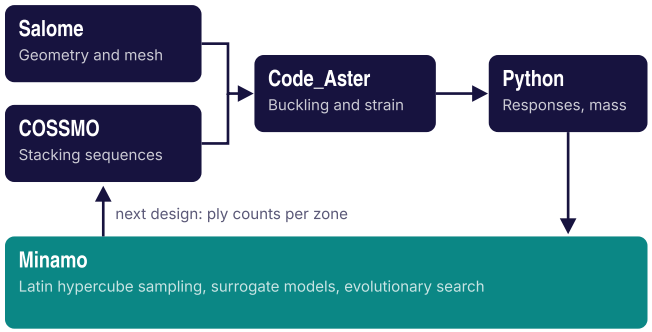
Load path

Stringers and hoop frames end at 8 plies each, stringers down from 12.

Tools	Salome, Code_Aster, COSSMO, Minamo, Python.
Compute	Lucia HPC cluster, batch mode from shell scripts. MPI tested: near-linear speed-up with CPUs.
Role	Structured and tested the whole computational chain, with full autonomy.

Automated pipeline

Open-source FE tools with Cenaero's COSSMO and Minamo, scripted end to end and run in batch on the Lucia HPC cluster.



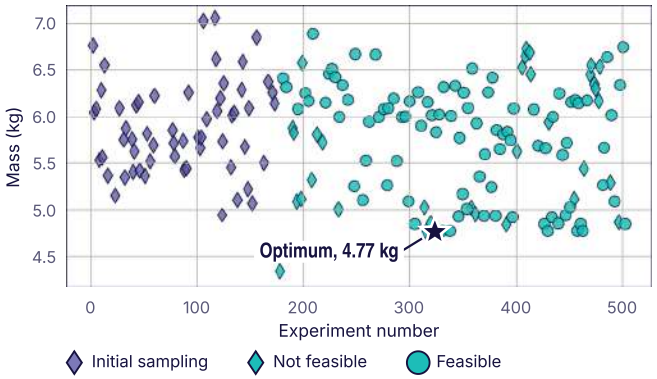
15 to 40 min
per enriched-model run, no MPI parallelisation

135 / 195
first sampling, simplified / enriched model

100
optimisation iterations

Surrogate-based optimisation

Surrogate models are rebuilt every iteration and searched by an evolutionary algorithm. 39 design variables.



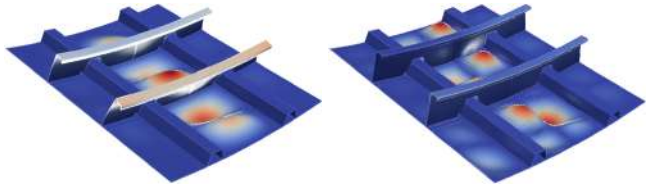
0
feasible designs in the first sampling

6.5 kg
first feasible design, roughly

4.77 kg
optimum, 5.2 % below the simplified model

Buckling analysis

Linear buckling in Code_Aster. The lowest mode of each zone is kept, and its load factor must stay above 1.



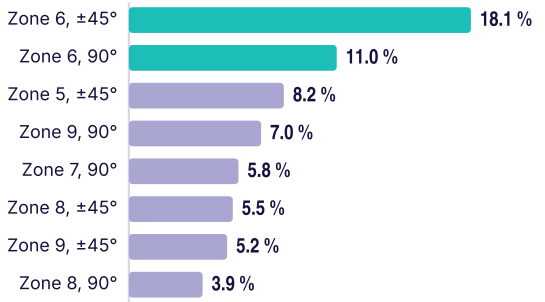
Load factor 2.29
12 plies in every zone

Load factor 4.27
Another mode, same layup

1. **Composite shells.** Skin, stringers and frames meshed in Salome, layups from COSSMO.
2. **15 constraints.** Buckling factor of each of 13 zones and of the whole panel, plus a strain reserve factor.
3. **Custom strain check.** Code_Aster has no maximum-strain criterion, so a Python script computes it.

Sensitivity analysis

Which ply counts drive the overall buckling factor, from the surrogate model of the simplified case.



Share of the overall buckling sensitivity. 19 more parameters: 35 %.

29 % **Two zone-6 parameters carry 29 %.** Zone 6 is on the load path, and it is where the optimiser added plies.

Honeycomb to carbon grid

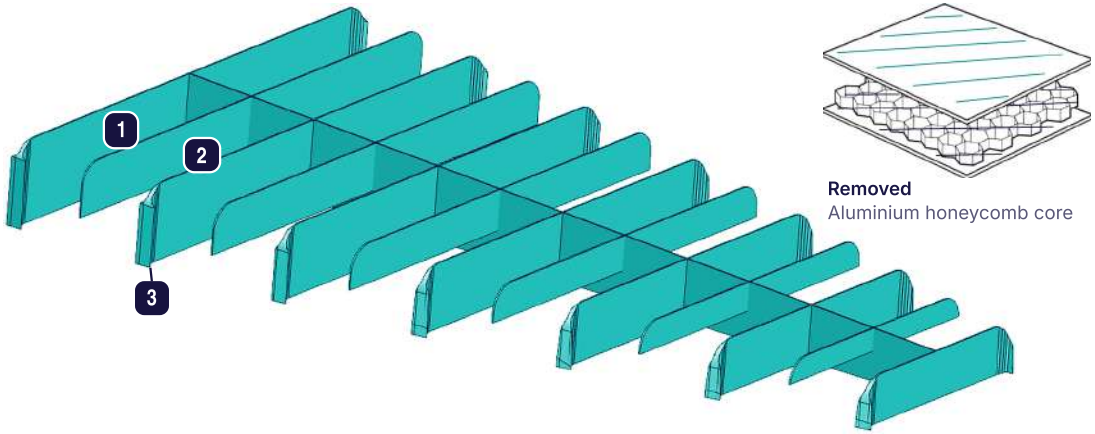
Airbus A350 flap support fairing top cover: honeycomb core replaced by a monolithic carbon grid

Internship at SABCA, Belgium. Aerospace company - February 2024 to July 2024

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Monolithic carbon grid

UD carbon ribs replace the aluminium honeycomb core. Same interfaces, so it drops in one-to-one.



Removed
Aluminium honeycomb core

1 Main ribs, 85 to 50 mm deep 2 Sub-ribs, 30 mm shallower 3 Joined directly to the bolted sides

Requirements and results

Mass	**−12.4 %** 2.51 kg, limit: not heavier
Buckling	**1.17** lowest load factor, limit above 1
Displacement	**4.4 mm** vertical, take-off, ultimate load, limit 5 mm
Bolts and joints	**2.6** reserve factor, limit above 1
Hail impact	**0.71 mm** peak deflection, honeycomb 1.02 mm

Tools Siemens NX with MSC Nastran: static, linear buckling and transient impact analyses, composite ply design, load cases from SABCA's model.

Grid layouts explored

Trial and error on layout, rib depth and thickness.

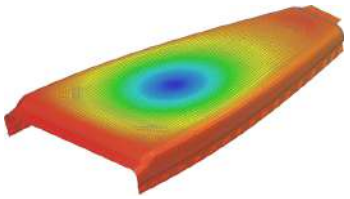


Transverse ribs | Longitudinal ribs | Isogrid-inspired

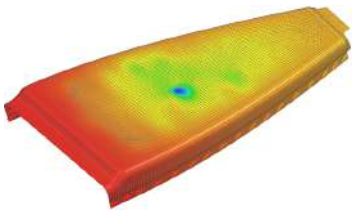
1. **First grid, 1.5 mm thick, 30 mm deep.** Seven transverse ribs already cut the vertical deflection drastically.
2. **Depth beats thickness.** Double depth at half thickness, same mass, deflected less. At 80 mm deep and 1 mm thick the ribs distort.
3. **Direct joints to the bolted sides** reduced the in-plane displacement in the ground run.
4. **Sub-ribs** flattened the local deflection peaks.

Hail strike and buckling

Transient impact and linear buckling in NX Nastran.

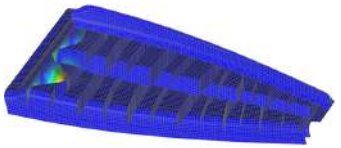


Honeycomb
Wide dent, peak 1.02 mm



Grid
Local dent, peak 0.71 mm

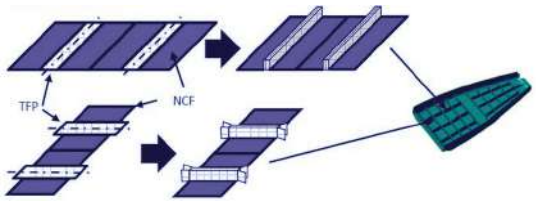
Same worst-case impact, 0.2 kg at 300 m/s. Each plot on its own colour scale.



0.44 → 1.47
buckling load factor in take-off, before and after reinforcing the front ribs

Production concept

Worked out with SABCA for their own facilities.



1. **TFP** stitches the UD carbon ribs into non-crimp fabric with a robotic gantry.
2. **HP-RTM** moulds the cover in one shot.
3. **Near net shape** keeps trimming and waste low. No honeycomb or foam core.

Monolithic carbon can be recycled, a honeycomb sandwich rarely is.

4-axis stone carver

Desktop CNC for marble statues and reliefs, with the block turning on a rotary axis inside a wet tub

Personal workshop project. Own design and build - September 2026 to present

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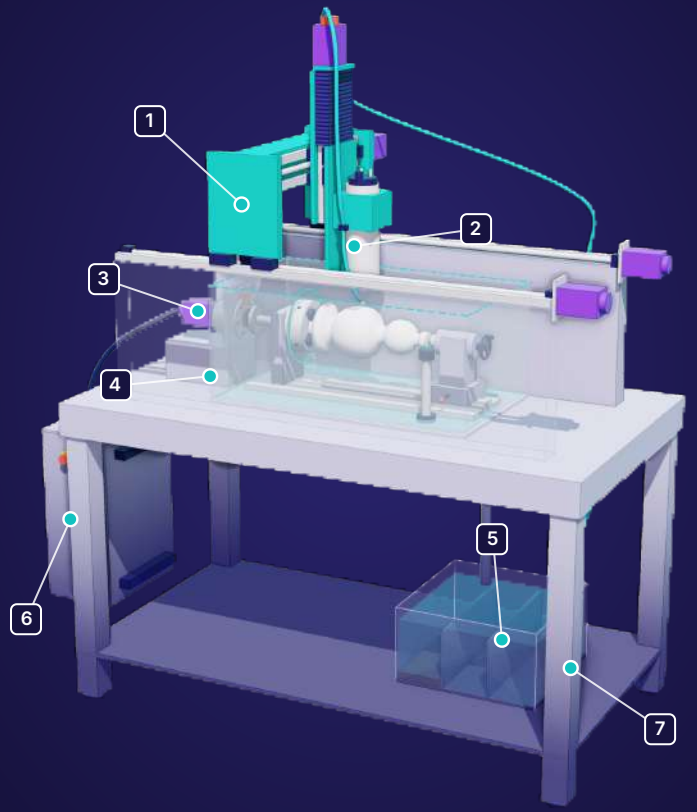
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Machine layout

A gantry over a wet tub on a steel stand, with the settling tank below and the control cabinet on the dry end.



Role Solo: concept, 3D layout, component selection, build

Stage Layout v0 done, build in progress

Components

Chosen for a 15 kg marble blank. Numbers match the layout.

1. **Gantry and axes.** Aluminium gantry, HGR20 rails, SFU1605 ball screws, twin closed-loop NEMA 23 on X, braked Z.
2. **Spindle.** 2.2 kW, 24,000 rpm, water-cooled, ER20, with a water nozzle and a sealed tool setter.
3. **Rotary A axis.** NEMA 23 with a 50:1 strain-wave reducer and disc brake, 4-jaw chuck and live-centre tailstock.
4. **Wet tub.**
5. **Settling tank.** Three filter chambers; the pump draws from the cleanest one.
6. **Control cabinet.** Drivers, VFD and a 30 mA RCD on the dry end, drip loops on every cable.
7. **Stand.** Steel box section.

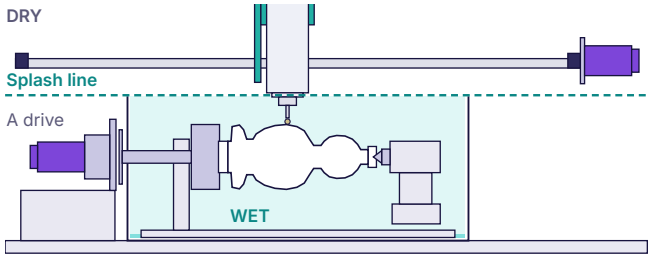
Envelope and targets

Design values

Travel	**400 × 250 × 200 mm**	X, Y, Z
Rotary A	**Ø200 mm**	swing, 300 mm between centres
Spindle	**2.2 kW**	24,000 rpm, water-cooled
A torque	**20 N·m**	minimum, 50:1 strain-wave
Deflection	**0.05 mm**	max. at the tool under 50 N
Blank	**15 kg**	marble, 140 × 140 × 280 mm

Control LinuxCNC on a Mesa 7i96S card.

Wet and dry zones



1. **Dry motion.** Rails, screws and motors ride on the side walls, above the splash line.
2. **Dry A drive.** Reducer and brake sit outside the tub; only a sealed shaft goes in.
3. **Two water loops.** Spindle coolant never meets the slurry.

Shapes and materials

Wet for stone, dry for wood.



Stone carving
A axis indexes the carved faces



Marble sculpting
A axis turns while carving



Wood panel
Flat 3-axis bed, cut dry

Materials Marble, limestone and alabaster cut wet with diamond tools; wood cut dry, with the water off.

LockDry

Neck-mounted hoodie clip with magnetic visor attachment

KU Leuven, Virtual Product Development. Individual course project - 2024

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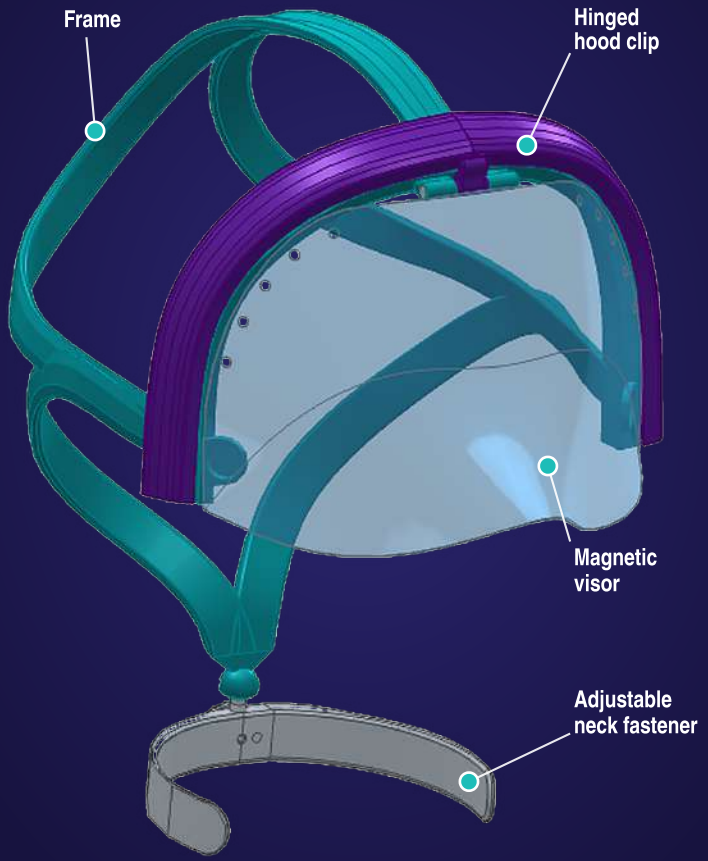
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Product design

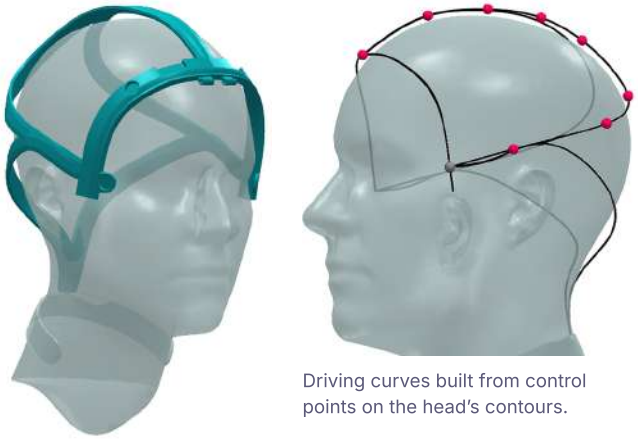
Concept, CAD, analysis and manufacturing plan, end to end.
Four modules, three joint types.



CAD & FEA	SolidWorks, Siemens NX
Making	Injection molding, CNC machining, press fitting
Prototype	3D-printed sample for shape and basic testing.

Ergonomics

Shaped on a reference head with rounded inner edges. Ball-and-socket neck joint for comfort, neck fastener adjusts 10 mm.



Driving curves built from control points on the head's contours.

User interface

Hinged hood clip, magnetic visor and a neck fastener adjusted with a gear wheel.



Hood clip and visor
Hinge opens, magnets hold.



Neck fastener
Rack and pinion, 10 mm range.

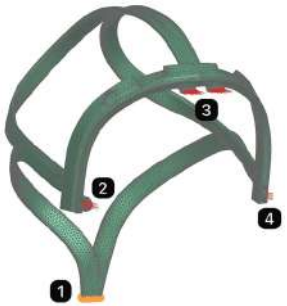




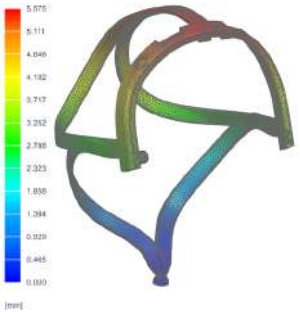
Exploded view
Four parts, assembly sequence defined.

Mechanical FEA for mass production

Siemens NX Nastran on the ABS frame under a worst-case wind load on the visor.



Loads and constraints



Displacement plot, ball-and-socket joint

50 km/h
wind load, worst case

68 g
ABS frame

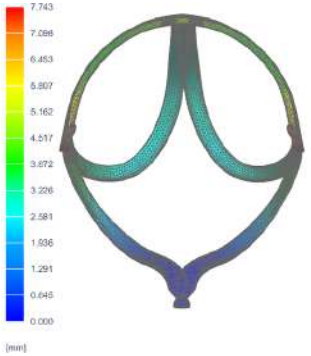
3
design iterations

Injection molding

The front ring blocks a straight pull. Concept: side slides, gate at the neck, demolding checked in FEA.



Mold concept
1 core, 2 side slides, 3 gate.



Demolding deformation in FEA
Stress stays below yield.
Buckling factor 5.3.

Muffler design

Three exhaust muffler concepts for 20 to 2000 Hz, simulated in Siemens NX and checked against theory

KU Leuven, Noise Abatement in Machines. Team of two - 2025

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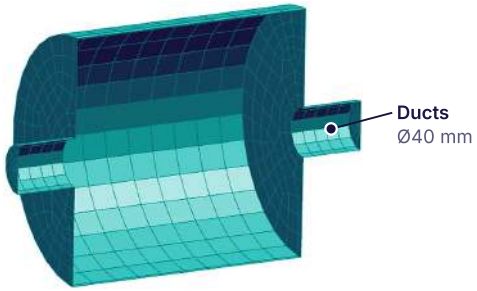
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From benchmark to three concepts

Exhaust muffler for an automotive company: 40 mm inlet and outlet, a 300 × 200 × 200 mm box, 20 to 2000 Hz, with the emphasis on low frequencies.

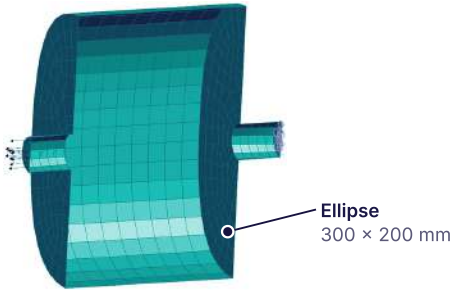


Benchmark

Circular expansion chamber, 200 × 200 mm. A 300 mm chamber does no better (16.2 dB) and has more surface area.

16.3 dB

average TL, the baseline for every step

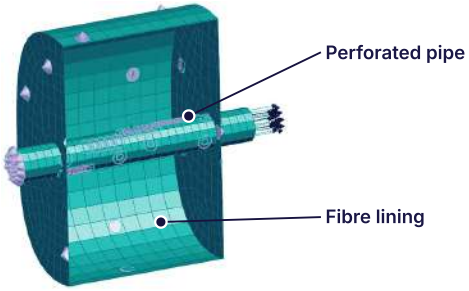


Reflective

Ellipse with a 90° outlet offset. Fewer axial standing waves, and more TL from 1309 to 1790 Hz at no extra mass.

Over 10 %

more TL than the benchmark on average, up to 1309 Hz

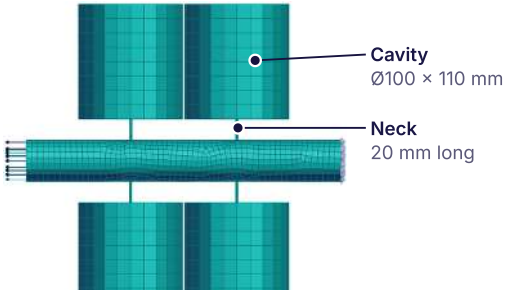


Absorptive

Perforated pipe and fibre lining in the ellipse: 3.2 mm holes at 10.5 mm spacing, lining on the Delany-Bazley-Miki model.

More than 2×

the benchmark TL up to 1700 Hz. 34 to 67 dB over the whole range



Meta-muffler

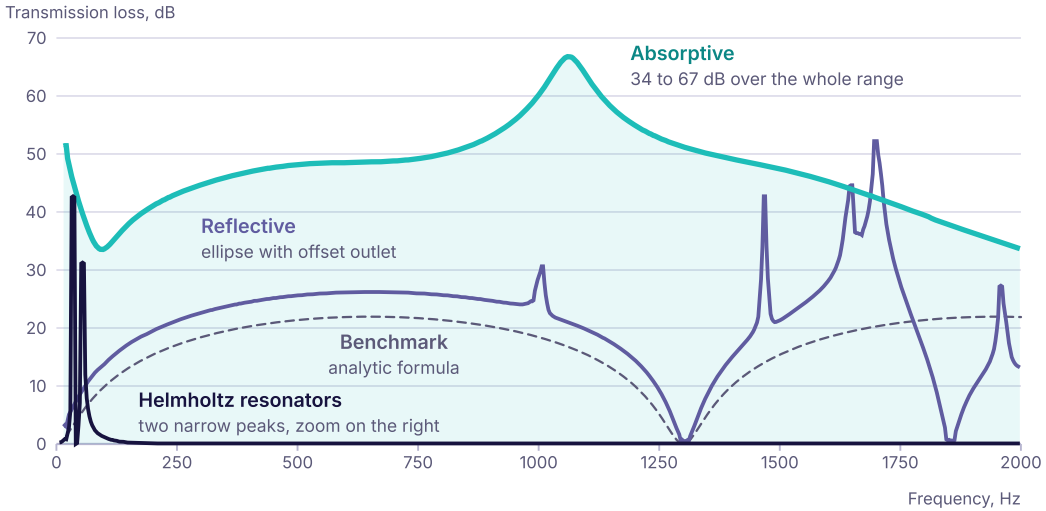
Four Helmholtz resonators on the duct, tuned to 40 and 60 Hz. Surface area 0.201 m² against 0.249 m² for the ellipse.

42 dB

peak TL at 37 Hz in BEM, 49 dB by transfer matrix. One resonator alone: 28 dB

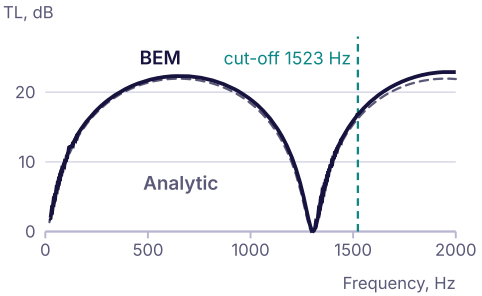
Transmission loss, 20 to 2000 Hz

TL is the sound power lost across the muffler. Dashed: the benchmark, from the analytic formula.

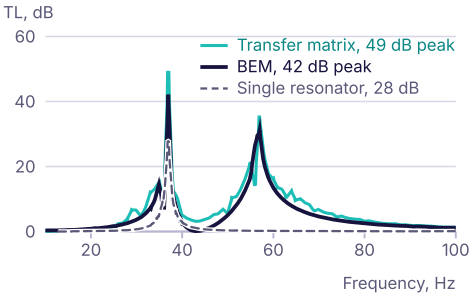


Checked against theory

BEM against the analytic formula, resonators against the transfer matrix.



Benchmark. BEM follows the analytic TL up to the 1523 Hz cut-off, where plane-wave theory ends.



Resonators. Tuned to 40 and 60 Hz, both methods peak at 37 and 57 Hz.

Tools Siemens NX with Simcenter 3D Acoustics BEM, 20 mm elements, finer than 10 per wavelength at 2000 Hz. Shell modes in Simcenter Nastran.

Shell 1 mm steel. Against titanium or a 1.5 mm wall, TL and radiated power moved by less than 1 %.

Aerial refuelling hose

Material and process selection for a 30 m probe-and-drogue refuelling hose, from 4249 candidates to one

KU Leuven, Materials Selection. Individual course project - 2025

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From 4249 materials to one

Screening and Ashby ranking in Granta EduPack, then shape, process, cost and eco-audit. Bars on a log scale.

4249

Materials in the Granta EduPack database

1313

Pass screening: −40 to 70 °C, 14.3 % elongation to rewind

25

Seven material indices: drag, pressure, flaws, vibration

15

Two weighted performance indices: TPEs and rubbers remain

5

Shortlist of five by density, from both subclasses

3

Can be made in the hose shape: TPV, natural rubber, silicone

1

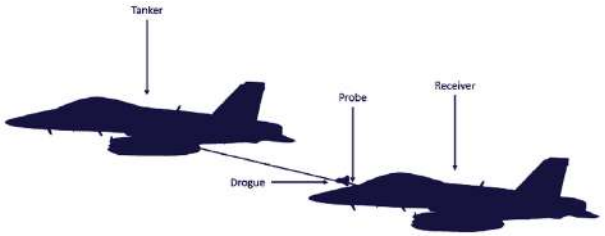
Natural rubber with carbon black, plus protective coatings

Tool Granta EduPack: Ashby charts, Eco Audit.

Method Indices derived from analytic load models, weighted performance indices, cost model.

Loads and limits

Worst-case numbers that turned the brief into screening limits and material indices.



12.3 kN
worst-case drag, 150 m/s at low altitude, 30 m hose

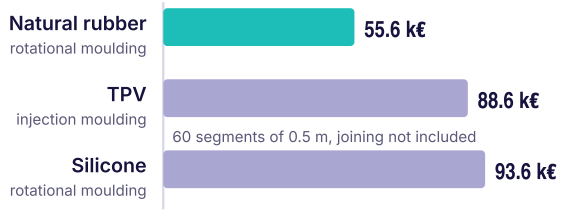
2 to 3.5 bar
fuel pressure inside the hose

−40 to 70 °C
service range, from high altitude to hot-day heating

14.3 %
elongation at break needed to rewind on a 0.45 m radius

Process and cost

Shape rules out two of the five. Cost model for 200 hoses, 30 m long each.



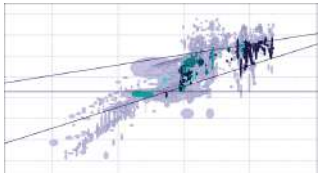
Rotational moulding makes the whole hose in one piece.

2.5 to 6 mm
wall thickness of rotational moulding

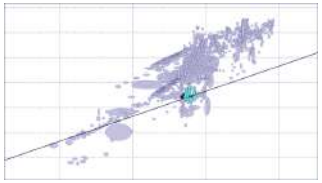
91 kg
silicone hose at the 6 mm wall, still within yield

Ashby ranking

Seven material indices from the loads, then two weighted performance indices.



Yield strength against density
Stress limit and specific strength



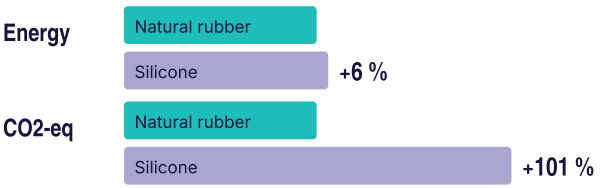
Weighted performance index
15 materials survive

Weights in the performance index

Mass	**0.5**	lightest wall under drag and pressure
Flaw tolerance	**0.3**	a crack through the wall stays stable
Vibration	**0.2**	stiffness and damping in turbulence

Eco audit and choice

Natural rubber against silicone, shipped by container ship and burnt at end of life.



Life-cycle total, relative to natural rubber.

1. **Best on the main index**, specific strength (MI1).
2. **Lowest batch cost**, 56 k€ against 94 k€ for silicone.
3. **Lower footprint** on energy and CO2. Only silicone lists aerospace among its uses.